

Introduction to Artificial Intelligence

Artificial Intelligence (AI) is a field of computer science focused on creating systems that can perform tasks that normally require human intelligence. These tasks can include understanding information, recognizing patterns, making predictions, solving problems, and supporting decisions.

Machine Learning

Machine Learning (ML) is a subset of Artificial Intelligence. Instead of explicitly programming every rule, an ML system learns patterns from data and uses those patterns to make predictions or decisions.

Types of Machine Learning

1. **Supervised Learning:** The model learns from labeled examples. Classification and regression are common supervised learning tasks.
2. **Unsupervised Learning:** The model works with data without predefined labels and attempts to discover useful patterns or groups.
3. **Reinforcement Learning:** An agent learns by interacting with an environment and receiving rewards or penalties for its actions.

Deep Learning

Deep Learning is a subfield of Machine Learning that uses neural networks with multiple layers. These networks can learn complex patterns from large amounts of data.

Natural Language Processing

Natural Language Processing (NLP) is an area of AI that deals with human language. Examples include text classification, translation, question answering, and summarization.

Key Idea

AI is the broader field, Machine Learning is a subset of AI that learns from data, and Deep Learning is a subset of Machine Learning that commonly uses multi-layer neural networks.